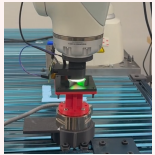
(a) Register real contacts



contact state q^* is read from the robot pose

kinematic, mounting and placement errors bias q^*; redo for each session or sensor

project mesh at q^*



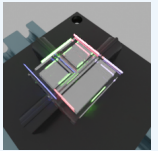
real image



registered label

cost: one registered pose per label

(b) Simulate contacts



mesh gives free geometry; appearance must be rendered

camera, LEDs, gel and contact model calibrated per sensor, or a renderer fitted on real pairs

render



sim depth (label)



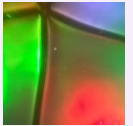
PBR



learned

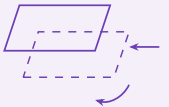
cost: calibration or real pairs; appearance gap remains

(c) AlignTac (ours): align the mesh to the real image



unlabeled real image, fixed

learned from the K registered contacts by perturbing their q^*



aligner $\mathcal{A}$ moves the mesh state q

align



mesh at teacher pose



aligned mesh = label

cost: K registered contacts; the rest unlabeled